

$$\exists A^{-1} \Leftrightarrow \det A \neq 0$$

$$A^{-1} = \frac{1}{\det A}$$

$$a_{ij} = (-1)^{i+j} \cdot i A_j$$

$$A = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \Rightarrow \det A = -1 + 1 - 1 = -1 \neq 0$$

$$A^D = \begin{bmatrix} \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} & -\begin{vmatrix} -1 & 1 \\ 0 & -1 \end{vmatrix} & \begin{vmatrix} -1 & 1 \\ 0 & 1 \end{vmatrix} \\ -\begin{vmatrix} 0 & -1 \\ 1 & -1 \end{vmatrix} & \begin{vmatrix} 1 & -1 \\ 0 & -1 \end{vmatrix} & -\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \\ \begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} \end{bmatrix} = \begin{bmatrix} -2 & -1 & -1 \\ -1 & -1 & -1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} -2 & -1 & -1 \\ -1 & -1 & -1 \\ 1 & 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

$$[A | I] \xrightarrow[\text{wierszowe}]{\text{op.}} [I | A^{-1}]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{w_2 + w_1} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{w_3 - w_2}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & -1 & -1 & 1 \end{array} \right] \xrightarrow{w_1 - w_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 1 & -1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & -1 & -1 & 1 \end{array} \right] \rightarrow$$

$$\xrightarrow{w_3 \cdot (-1)} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 1 & -1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & -1 \end{array} \right]$$

Zad. 6

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det A} \cdot [A^D]^T = \frac{1}{7} \cdot \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

$$\det A = 3 \cdot 2 - 1 \cdot (-1) = 7 \neq 0$$

$$A^D = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$$

$$[A^D]^T = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

$$\begin{aligned} \det B &= 1 \cdot 2 \cdot 2 + 3 \cdot 1 \cdot 3 + 1 \cdot 2 \cdot 1 \\ &\quad - 3 \cdot 2 \cdot 1 - 1 \cdot 1 \cdot 1 - 2 \cdot 2 \cdot 3 = \\ &= 4 + 9 + 2 - 6 - 1 - 12 = \\ &= 15 - 7 - 12 = -4 \neq 0 \end{aligned}$$

$$B^D = \begin{bmatrix} \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} & -\begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} & \begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} \\ -\begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} & \begin{vmatrix} 1 & 3 \\ 1 & 2 \end{vmatrix} & -\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} \\ \begin{vmatrix} 2 & 3 \\ 2 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ 3 & 2 \end{vmatrix} \end{bmatrix} = \begin{bmatrix} 3 & -5 & 1 \\ -1 & -1 & 1 \\ -4 & 8 & -4 \end{bmatrix}$$

$$(B^D)^T = \begin{bmatrix} 3 & -1 & -4 \\ -5 & -1 & 8 \\ 1 & 1 & -4 \end{bmatrix}$$

$$B^{-1} = -\frac{1}{4} \begin{bmatrix} 3 & -5 & 1 \\ -1 & -1 & 1 \\ -4 & 8 & -4 \end{bmatrix} = \begin{bmatrix} -\frac{3}{4} & \frac{5}{4} & -\frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & -\frac{1}{4} \\ 1 & -2 & 1 \end{bmatrix}$$

tw. Laplace'a

$$\begin{vmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & -2 & -1 \\ 2 & 1 & 0 & -1 \\ 3 & 1 & 1 & 0 \end{vmatrix} = 1 \cdot (-1)^{1+1} \begin{vmatrix} 1 & -2 & -1 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \end{vmatrix} + 0 \cdot (-1)^{1+2} \begin{vmatrix} 1 & 2 & 4 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \end{vmatrix} +$$

$$+ 2 \cdot (-1)^{1+3} \begin{vmatrix} 1 & 2 & 4 \\ 1 & -2 & -1 \\ 1 & 1 & 0 \end{vmatrix} + 3 \cdot (-1)^{1+4} \begin{vmatrix} 1 & 2 & 4 \\ 1 & -2 & -1 \\ 1 & 0 & -1 \end{vmatrix} =$$

$$\begin{vmatrix} 1 & 1 & 2 & 4 \\ 0 & -1 & -2 & -1 \\ 2 & 1 & 0 & -1 \\ 3 & -1 & 1 & 0 \end{vmatrix} \stackrel{\substack{w_3 - 2w_1 \\ w_4 - 3w_1}}{=} \begin{vmatrix} 1 & 1 & 2 & 4 \\ 0 & -1 & -2 & -1 \\ 0 & -1 & -4 & -9 \\ 0 & -4 & -5 & -12 \end{vmatrix} = 1 \cdot (-1)^{1+1} \begin{vmatrix} 1 & -2 & -1 \\ -1 & -4 & -9 \\ -4 & -5 & -12 \end{vmatrix} =$$

$$= \begin{vmatrix} 1 & -2 & -1 \\ -1 & -4 & -9 \\ -4 & -5 & -12 \end{vmatrix}$$

$$\begin{vmatrix} -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 \\ -3 & 1 & 2 & 0 \\ 1 & 4 & 3 & 1 \end{vmatrix} = -2 \cdot (-1)^{1+1} \begin{vmatrix} 3 & 0 & 0 \\ 1 & 2 & 0 \\ 4 & 3 & 1 \end{vmatrix} = -2 \begin{vmatrix} 3 & 0 & 0 \\ 1 & 2 & 0 \\ 4 & 3 & 1 \end{vmatrix} =$$

$$= -2(6 + 0 + 0 - 0 - 0 - 0) = -12$$

$$\begin{vmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & -2 & 2 \\ 1 & 1 & 1 & -2 \\ 0 & 1 & 2 & 1 \end{vmatrix} \stackrel{w_3 - w_1}{=} \begin{vmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & -2 & 2 \\ 0 & -1 & 2 & -2 \\ 0 & 1 & 2 & 1 \end{vmatrix} =$$

$$= 1 \cdot (-1)^{1+1} \begin{vmatrix} 1 & -2 & 2 \\ -1 & 2 & -2 \\ 1 & 2 & 1 \end{vmatrix} = \cancel{2} - \cancel{4} + \cancel{4} - \cancel{4} + \cancel{4} - \cancel{2} = 0$$

$$\left| \begin{array}{cccc|c} 1 & 1 & 1 & 1 & w_2 - w_1 \\ 1 & 2 & 3 & 4 & = \\ 1 & 4 & 9 & 16 & w_3 - w_1 \\ 1 & 8 & 27 & 64 & w_4 - w_1 \end{array} \right| = \left| \begin{array}{cccc|c} 1 & 1 & 1 & 1 & \\ 0 & 1 & 2 & 3 & \\ 0 & 3 & 8 & 15 & \\ 0 & 7 & 26 & 63 & \end{array} \right| = 1 \cdot (-1)^{1+1} \left| \begin{array}{ccc} 1 & 2 & 3 \\ 3 & 8 & 15 \\ 7 & 26 & 63 \end{array} \right| =$$

$$= \left| \begin{array}{ccc|c} 1 & 2 & 3 & w_2 - 3w_1 \\ 3 & 8 & 15 & \\ 7 & 26 & 63 & w_3 - 7w_1 \end{array} \right| = \left| \begin{array}{ccc|c} 1 & 2 & 3 & \\ 0 & 2 & 6 & \\ 0 & 12 & 42 & \end{array} \right| = 1 \cdot (-1)^{1+1} \left| \begin{array}{cc} 2 & 6 \\ 12 & 42 \end{array} \right| =$$

$$= 84 - 72 = 12$$

$$\left| \begin{array}{cccc|c} 0 & 1 & 2 & 3 & w_3 - 2w_2 \\ -1 & 0 & 1 & 2 & w_4 - 3w_2 \\ -2 & -1 & 0 & 1 & \\ -3 & -2 & -1 & 0 & \end{array} \right| = \left| \begin{array}{cccc|c} 0 & 1 & 2 & 3 & \\ -1 & 0 & 1 & 2 & \\ 0 & -1 & -2 & -3 & \\ 0 & -2 & -4 & -6 & \end{array} \right| = -1 \cdot (-1)^{1+2} \left| \begin{array}{ccc} 1 & 2 & 3 \\ -1 & -2 & -3 \\ -2 & -4 & -6 \end{array} \right| =$$

$$= 1 \cdot \left| \begin{array}{ccc} 1 & 2 & 3 \\ -1 & -2 & -3 \\ -2 & -4 & -6 \end{array} \right|_{w_2 \leftrightarrow w_1} = 1 \cdot \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

$$2^0 \left[\begin{array}{cccc} 1 & 2 & 3 & -1 \\ 0 & -1 & -2 & 0 \\ 1 & 1 & 1 & -1 \end{array} \right] \xrightarrow{w_3 - w_1} \left[\begin{array}{cccc} 1 & 2 & 3 & -1 \\ 0 & -1 & -2 & 0 \\ 0 & -1 & -2 & 0 \end{array} \right] \rightarrow$$

$$\rightarrow \left[\begin{array}{cccc} 1 & 2 & 3 & -1 \\ 0 & -1 & -2 & 0 \end{array} \right]$$

$$\text{Kości schodkowe} = \text{rz } A = 2$$

Rząd - wymiary maksymalnej podmacierzy o wyznaczniku $\neq 0$
- maks ilość wektorów liniowo niezależnych

$\det A \neq 0 \Rightarrow A$ jest liniowo niezależna $\Rightarrow \text{rz } A = n$

$$\begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & -1 & -2 & 0 \\ 1 & 1 & 1 & -1 \end{bmatrix}_{3 \times 4} \quad \text{rz } A = \min\{3, 4\} = 3$$

Sprawdzamy, czy $\text{rz } A = 3$?

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 1 & 1 & 1 \end{vmatrix} = 0 \quad \begin{vmatrix} 1 & 2 & -1 \\ 0 & -1 & 0 \\ 1 & 1 & -1 \end{vmatrix} = 0$$

$\text{rz } A \neq 3$

$$\begin{vmatrix} 2 & 3 & -1 \\ -1 & -2 & 0 \\ 1 & 1 & -1 \end{vmatrix} = 0 \quad \begin{vmatrix} 1 & 3 & -1 \\ 0 & -2 & 0 \\ 1 & 1 & -1 \end{vmatrix} = 0$$

Sprawdzamy czy $\text{rz } A = 2$?

$$\begin{vmatrix} 1 & 2 \\ 0 & -1 \end{vmatrix} = -1 \neq 0$$

$\Downarrow$
 $\text{rz } A = 2$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix} = [\underline{1} \ 1 \ 1] \quad \text{rk } A = 1$$

$$B = \begin{bmatrix} 2 & 1 & 3 & 6 \\ 1 & 0 & 1 & 2 \\ 3 & 1 & 4 & 8 \end{bmatrix} \xrightarrow[2w_3 - 3w_1]{2w_2 - w_1} \begin{bmatrix} 2 & 1 & 3 & 6 \\ 0 & -1 & -1 & -2 \\ 0 & -1 & -1 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 3 & 6 \\ 0 & -1 & -1 & -2 \end{bmatrix}$$

$\text{rk } A = 2$

$$C = \begin{bmatrix} 1 & -1 & 1 & 2 \\ 2 & -2 & 2 & 4 \\ 1 & 1 & 2 & 1 \\ 2 & 0 & 3 & 3 \end{bmatrix} \xrightarrow{w_2 = w_1 \cdot 2} \begin{bmatrix} 1 & -1 & 1 & 2 \\ 1 & 1 & 2 & 1 \\ 2 & 0 & 3 & 3 \end{bmatrix}$$